Neonatal Oral Antibiotics after Intravenous at Home (NOAH approach)

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Authors: **Lesley Jackson**
Neonatal Consultant NHS GGC
and NHS GGC Neonatal Hospital at Home Service

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NHS GGC Document Development Coversheet*

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Author	Lesley Jackson		
Executive Lead			

Proposed groups to present document to:		
GGC Consultants	GGC Neonatal DSCNs	Pharmacy RHC , PRM , RAH

Date	**Version**	**Group**	**Reason**	**Outcome**
20/05/2026	1.0	RHC Neonatal Consultants & GGC DSCNs, pharmacy Teams	PI, FIO	

Examples of reasons for presenting to the group	**Examples of outcomes following meeting**
• Professional input required re: content (PI)	• Significant changes to content required – refer to Executive Lead for guidance (SC)
• Professional opinion on content (PO)	• To amend content & re-submit to group (AC&R)
• General comments/suggestions (C/S)	• For minor revisions (e.g. format/layout) – no need to re-submit to group (MR)
• For information only (FIO)	• Recommend proceeding to next stage (PRO)
• For proofing/formatting (PF)	• For upload to Intranet (INT)
• Final Approval (FA)	• Approved (A) or Not Approved, revisions required (NARR)

***To be attached to the document under development/review and presented to the relevant group**

Please record details of any changes made to the document in the table below

Date	Record of changes made to document

Neonatal Oral Antibiotics at Home after IV (NOAH approach)
Standard Operating Procedure

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1. Introduction

This policy applies to all patients that are admitted to NHS Greater Glasgow & Clyde Neonatal Services including Neonatal Hospital at Home Services and is to be followed by all nursing and medical staff within the Neonatal Services and relevant associated Board staff

2. Aim: To ensure safe and effective delivery of oral antibiotics at home following a switch from IV therapy to oral (IVOST) utilising a standardised process and care delivered at home by the Neonatal H@H Team

3. Policy Statement

The clinical priority is ensuring that babies meeting the clinical criteria for NOAH are identified and parents offered this care at home. This care is then delivered in a consistent and safe manner on the postnatal wards and in the family home.

4. Definitions

Admission to NOAH pathway: a patient/baby who is identified as requiring a prolonged course of antibiotics , who is otherwise well with negative blood cultures and receiving care on the postnatal wards in Princess Royal Maternity, the Royal Alexandra Hospital, or the Royal Hospital for Children

5. Standard Management

Currently in the UK, standard management of suspected early-onset neonatal infection involves a minimum of 36 hours of intravenous (IV) antibiotics. Antibiotics are typically stopped after 36 - 48 hours if the baby is well, CRP results are reassuring, and any cultures taken are negative. This follows NICE guidance.

However, when there are ongoing concerns for infection and/or a raised CRP, IV antibiotics are continued for up to 7 days. Use of IV antibiotics has a number of disadvantages including increased duration of inpatient stay and multiple cannulation attempts.

- Neonatal IV to oral antibiotics at home is often referred to as the NOAH approach and is an example of IV-to-oral switch therapy (IVOST)
- Aligns with evidence based from RAIN study and is established practice in many areas of NHS England

NOAH Dosing Regimen

- **Amoxicillin 30 mg/kg three times daily following initial treatment with IV cefotaxime or IV Benzylpenicillin and IV gentamicin depending on referral unit**
- **Total duration of antibiotics remains 7 days (IV and oral combined)**

- **Estimate 2-3 babies QEUH eligible /month.**

- **1-2 babies /month between PRM and RAH**
- **small numbers but potential to align 4-5 days per baby care days to home utilising this approach**
- **Positive impact on families and bonding , siblings**

Criteria based to be eligible for NOAH approach at home:

- Babies identified at risk of early onset Sepsis with raised CRP but negative BC at 48 hrs. Otherwise well receiving care on PNWs
- ≥37+0 weeks
- Clinically well, parents happy for treatment and care at home
- Falling CRP at time of acceptance onto NOAH pathway and BC negative
- Peak CRP not >100
- 48hrs of IV cefotaxime/ Benpen and Gent until BC negative
- Parents and referring clinical team happy for care to be continued at home
- IVOST Switch to oral amoxicillin 30mg/kg tid for total of 7 days
- Total of 7 days treatment with Antibiotics
- Refer to Neonatal H@H Team **07866957495**
- Capacity available on Neonatal H@H and admitted under Consultant on for H@H by Interface Admin Hub (alerts and automatic admission letter to primary care generated)
- Admitted onto Neonatal H@H Trak ward
- Written prescription by referring Team for oral amoxicillin pre discharge from PNW of referring unit (use dose banded Guidance, see table below)
- Prescribe a total of 7 days of Antibiotics , e.g. if 3 days IV received 4 days of oral
- Pathway for this agreed with Pharmacy
- Supply of oral amoxicillin 125mg/5ml in each NNU
- TTO packs (to take out packs)
- Labelled and dispensed by H@H Nurse on duty
- Parents trained to administer oral antibiotics (H@H team can do this)
- CRP taken by Neonatal H@H team first day at home (next working day following discharge from PNWs)
- CRP pre last day of treatment (day 6 of Antibiotic treatment)

- Daily clinical review by Neonatal H@H team including temperature, RR, HR, saturation. Review of feeding. Weight at hour 60 and thereafter as indicated. Ask and record any parental/ carer concerns
- Daily discussion with on call H@H consultant. Any concerns will generate repeat bloods and referral back to base unit for review
- Daily review documented on Badger
- Parents/carers given advice on contact processes out of hours. If relates to practical admission of antibiotics or sundries issue advice to phone NNU RHC level 2 . If relates to wellbeing contact RHC ED /NHS 111. GP informed of virtual ward status on admission by automatically generated specific letter
- Escalate to review in NNU if concerns , CRP not falling , any other concerns
- Data recorded on treatment days at home , any readmissions to Paediatric services in first 28 days of life
- At discharge, discharged from virtual ward. IDL to GP. Handover over to Health Visitor

Table 1 Dose Banding of Oral Amoxicillin 125mg/5ml to prescribe for Home

Amoxicillin 125mg/5ml oral suspension

select corresponding dose for weight band as per table

Infant Weight (kg)	Dose to Prescribe (mg)	Volume to Give (mL)	Frequency
2.0 – 2.39 kg	65 mg	2.6 mL	THREE times a day
2.4 – 2.79 kg	75 mg	3 mL	
2.8 – 2.99 kg	85 mg	3.4 mL	
3.0 – 3.29 kg	95 mg	3.8 mL	
3.3 – 3.49 kg	105 mg	4.2 mL	
3.5 – 4.0 kg	115 mg	4.6 mL	
4 kg and above	125 mg	5 mL	



Figure 1 Labelling of TTO Supplies of Oral amoxicillin 125mg/5ml

References

1. Keij F, Kornelisse R, Hartwig N et al.

   Efficacy and safety of switching from intravenous to oral antibiotics (amoxicillin–clavulanic acid) versus a full course of intravenous antibiotics in neonates with probable bacterial infection (RAIN): a multicentre, randomised, open-label, non-inferiority trial

   2.Malchau Carlsen EL, Dungu KHS, Lewis A*, et al*
   Switch from intravenous-to-oral antibiotics in neonatal probable and proven early-onset infection: a prospective population-based real-life multicentre cohort study
   Archives of Disease in Childhood - Fetal and Neonatal Edition 2024;**109:**34-40

3,Early onset sepsis in the neonate: prevention and treatment (1060) | NHSGGC accessed 20052026

4. Overview | Neonatal infection: antibiotics for prevention and treatment | Guidance | NICE accessed 200526